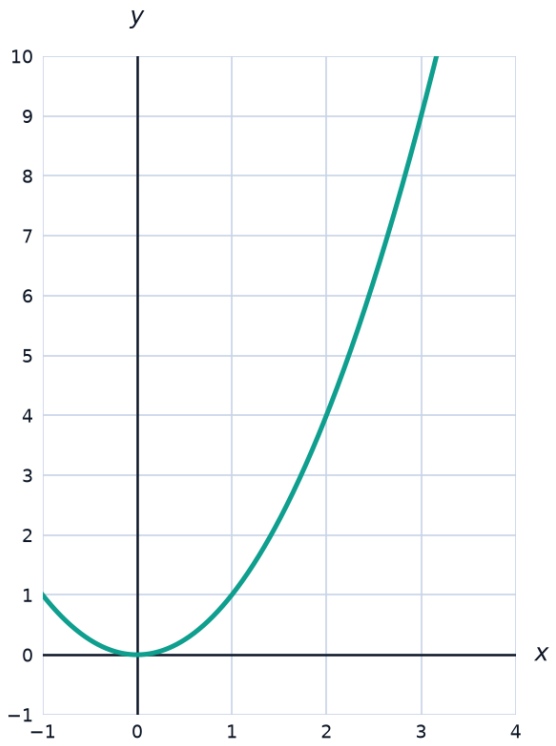
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Area under a curve**

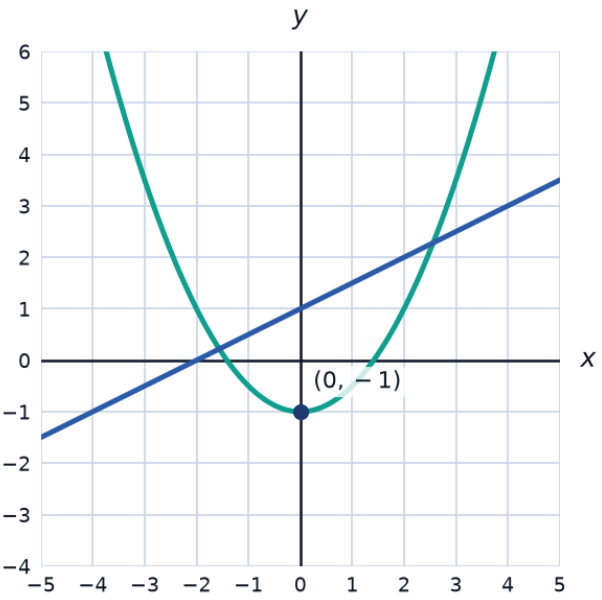
Find the area under $y = x^2$ from $x = 0$ to $x = 3$, using a definite integral.



© ClipSAT — clipsat.org | All Rights Reserved

2 [Foundational] • Target: Basic antiderivatives (the power rule)

Find $\int x^4\,dx$.



3 [Foundational] • Target: Basic antiderivatives (the power rule)

Find $\int 3x^2\,dx$.

4 [Foundational] • Target: Basic antiderivatives (the power rule)

Find $\int \sqrt{x}\,dx$.

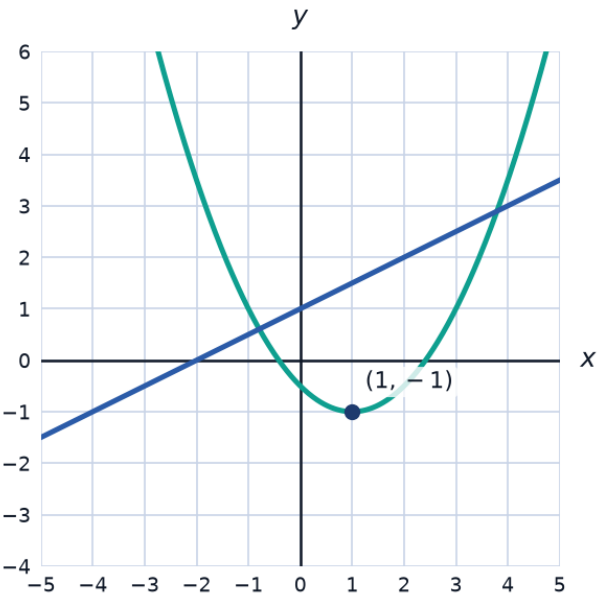
5 [Foundational] • Target: Basic antiderivatives (the power rule)

Find $\int \frac{1}{x^3}\,dx$.

© ClipSAT — clipsat.org | All Rights Reserved

6 [Foundational] • Target: Antiderivatives of polynomials

Find $\int (4x^3 - 6x + 5)\,dx$.



7 [Foundational] • Target: Antiderivatives of polynomials

Find $\int (2x + 7)\,dx$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Antiderivatives of polynomials

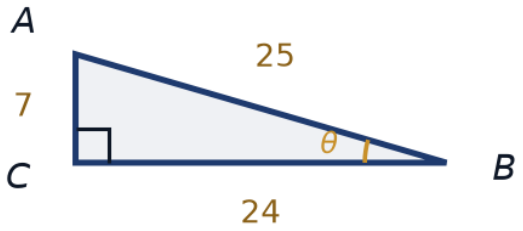
Find $\int (x^2 - 4x + 3)\,dx$.

9 [Intermediate] • Target: Antiderivatives of trigonometric and exponential functions

Find $\int \cos x\,dx$.

10 [Intermediate] • Target: Antiderivatives of trigonometric and exponential functions

Find $\int \sin x\,dx$.



11 [Intermediate] • Target: Antiderivatives of trigonometric and exponential functions

Find $\int e^x\,dx$.

© ClipSAT — clipsat.org | All Rights Reserved

12 [Intermediate] • Target: Evaluating definite integrals

Evaluate $\int_0^2 3x^2\,dx$.

13 [Intermediate] • Target: Evaluating definite integrals

Evaluate $\int_1^3 2x\,dx$.

14 [Intermediate] • Target: Evaluating definite integrals

Evaluate $\int_0^4 x\,dx$.

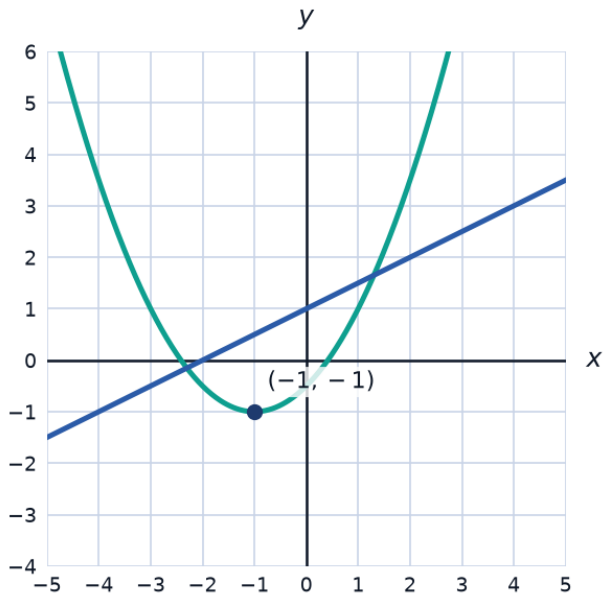
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Evaluating definite integrals • Cross-Lesson: 4.01 Trigonometry

Evaluate $\int_{-1}^2 3x^2\,dx$.

16 [Challenging] • Target: Definite integrals with polynomial integrands • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

Evaluate $\int_0^2 (x^2+1)\,dx$.



17 [Challenging] • Target: Definite integrals with polynomial integrands • Cross-Lesson: 6.01 Limits & Derivatives

Evaluate $\int_1^3 (2x^2-3x+1)\,dx$.

18 [Challenging] • Target: Definite integrals with polynomial integrands • Cross-Lesson: 8.01 Exponentials & Logarithms

Evaluate $\int_0^1 (6x^2-4x+2)\,dx$.

19 [Challenging] • Target: Definite integrals with polynomial integrands • Cross-Lesson: 9.01 Statistics & Probability

Evaluate $\int_{-2}^1 (x^2-x)\,dx$.

© ClipSAT — clipsat.org | All Rights Reserved

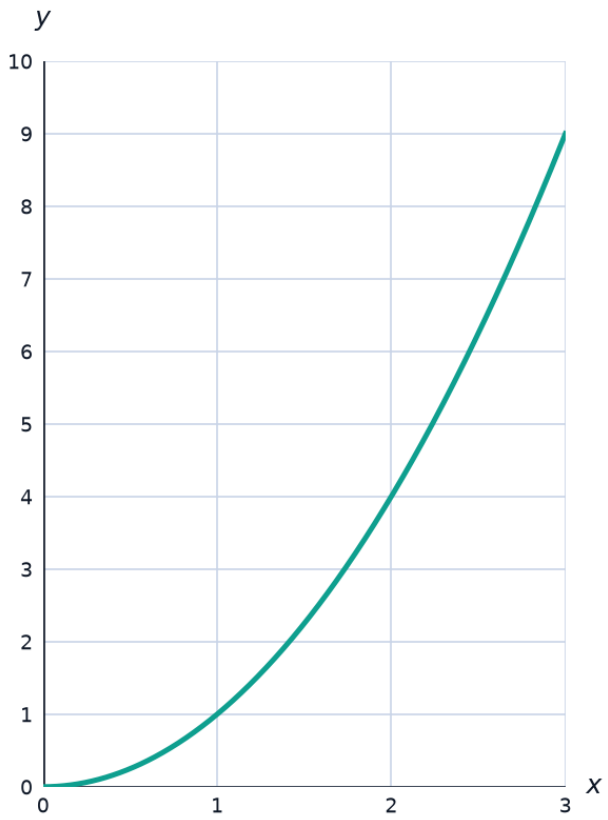
20 [Challenging] • Target: Definite integrals of trigonometric and exponential functions • Cross-Lesson: 1.01 Algebra & Equations

Evaluate $\int_0^{\pi/2} \cos x \, dx$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: MCQ — Area under a curve

The graph shows $y = x^2$. Find the area under the curve from $x = 0$ to $x = 3$, using a definite integral.



a. 9

b. 18

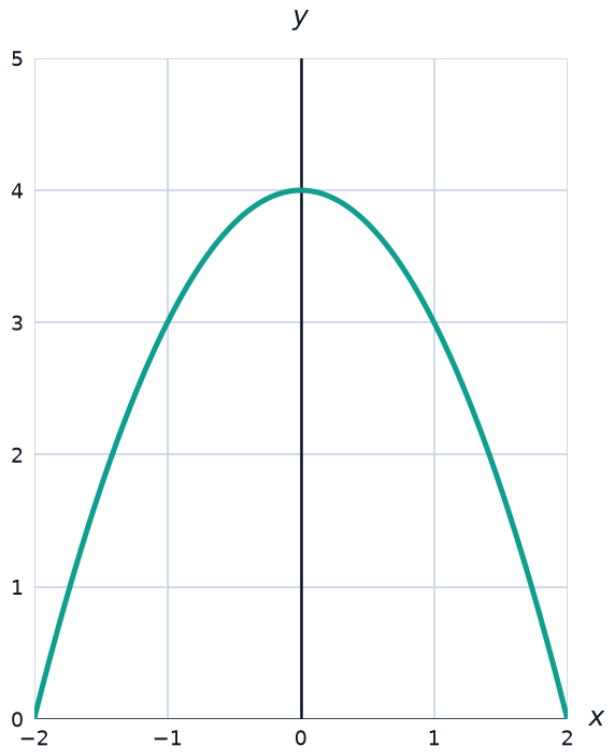
c. 3

d. 27

© ClipSAT — clipsat.org | All Rights Reserved

22 [Foundational] • Target: MCQ — Area under a curve

The graph shows $y = 4 - x^2$. Find the area under the curve from $x = -2$ to $x = 2$.



a. $\frac{16}{3}$

b. 32

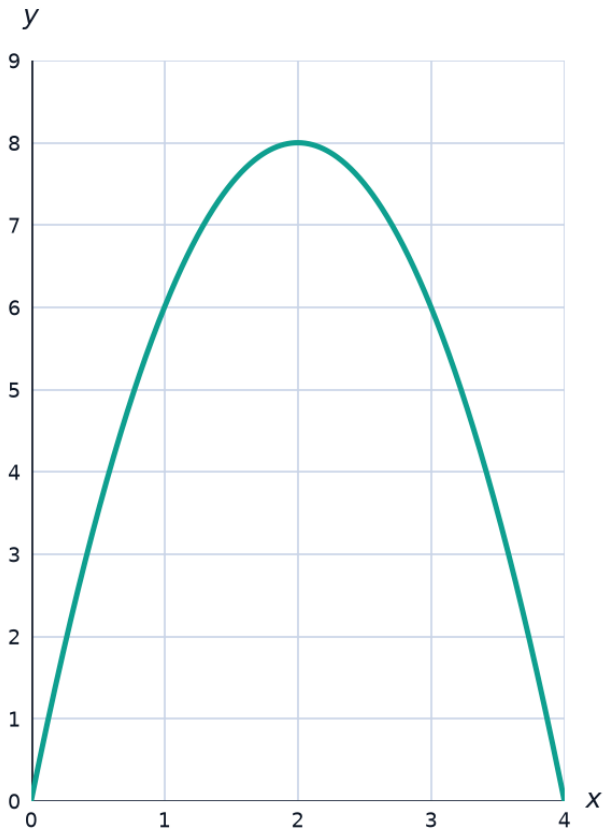
c. 16

d. $\frac{32}{3}$

© ClipSAT — clipsat.org | All Rights Reserved

23 [Foundational] • Target: MCQ — Word problems: integration in context

The graph shows a particle's velocity $v(t) = 8t - 2t^2$ for $0 \le t \le 4$. Find the particle's displacement (the area under the curve).



a. $\frac{64}{3}$

b. 16

c. 32

d. 64

24 [Foundational] • Target: MCQ — Basic antiderivatives (the power rule)

Find $\int x^6\,dx$.

a. $\frac{x^6}{7} + C$

b. $7x^5 + C$

c. $\frac{x^7}{6} + C$

d. $\frac{x^7}{7} + C$

25 [Foundational] • Target: MCQ — Basic antiderivatives (the power rule)

Find $\int 7\,dx$.

a. C

b. $7x + C$

c. $7 + C$

d. $\frac{7x^2}{2} + C$

© ClipSAT — clipsat.org | All Rights Reserved

26 [Foundational] • Target: MCQ — Antiderivatives of polynomials

Find $\int (x^2 - 6x + 5)\,dx$.

a. $\frac{x^3}{3} - 3x^2 + 5x + C$

b. $\frac{x^3}{3} - 6x^2 + 5x + C$

c. $\frac{x^3}{3} - 3x^2 + C$

d. $x^3 - 3x^2 + 5x + C$

27 [Foundational] • Target: MCQ — Antiderivatives of trigonometric and exponential functions

Find $\int (e^x + 2)\,dx$.

a. $e^x + 2 + C$

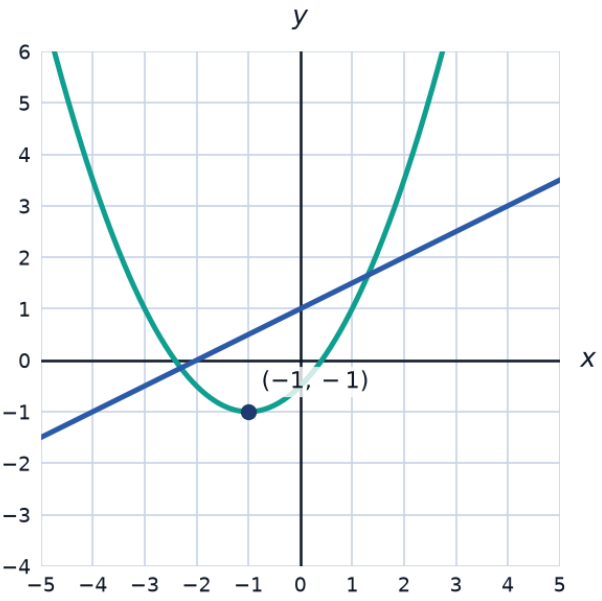
b. $e^x + 2x + C$

c. $xe^x + 2x + C$

d. $e^x + C$

28 [Foundational] • Target: MCQ — Evaluating definite integrals

Evaluate $\int_0^5 x\,dx$.



a. 5

b. 12.5

c. 25

d. 10

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

© ClipSAT — clipsat.org | All Rights Reserved

29 [Intermediate] • Target: MCQ — Definite integrals with polynomial integrands

Evaluate $\int_1^4 (3x^2 - 2x + 1)\,dx$.

a. 50

b. 52

c. 63

d. 51

30 [Intermediate] • Target: MCQ — Definite integrals of trigonometric and exponential functions

Evaluate $\int_0^{\pi} \cos x\,dx$.

a. π

b. -2

c. 0

d. 2

31 [Intermediate] • Target: MCQ — Signed area and regions below the axis

Evaluate $\int_0^3 (x - 3)\,dx$ and explain what the sign of the result means.

a. 4.5; the region lies above the x-axis

b. -4.5; the region lies above the x-axis

c. 4.5; the region lies below the x-axis

d. -4.5; the region lies below the x-axis

32 [Intermediate] • Target: MCQ — Properties of definite integrals

Given $\int_0^3 f(x)\,dx = 6$ and $\int_3^7 f(x)\,dx = 9$, find $\int_0^7 f(x)\,dx$.

a. 13

b. 3

c. 15

d. 54

33 [Intermediate] • Target: MCQ — Finding a function from its derivative

A function satisfies $f'(x) = 4x^2 + 3x$ and $f(0) = 2$. Find $f(x)$.

a. $\frac{4x^3}{3} + 3x^2 + 2$

b. $\frac{4x^3}{3} + 1.5x^2$

c. $\frac{4x^3}{3} + 1.5x^2 + 2$

d. $4x^3 + 1.5x^2 + 2$

© ClipSAT — clipsat.org | All Rights Reserved

34 [Intermediate] • Target: MCQ — Average value of a function

Find the average value of $f(x) = 3x + 2$ on $[0, 4]$.

a. 32

b. 4

c. 14

d. 8

35 [Intermediate] • Target: MCQ — Word problems: integration in context

A factory's production rate is $p(t) = 50 + 4t$ items per hour, for $0 \le t \le 8$. Find the total number of items produced during this 8-hour period.

a. 528

b. 656

c. 450

d. 400

36 [Intermediate] • Target: MCQ — Evaluating definite integrals

Find $\int -x^2\,dx$.

a. $-\frac{1}{3}x^3 + C$

b. $-\frac{1}{4}x^3 + C$

c. $-\frac{1}{2}x^2 + C$

d. $-x^3 + C$

37 [Intermediate] • Target: MCQ — Definite integrals with polynomial integrands

Find $\int (4x + 9)\,dx$.

a. $2x^2 + 9x + C$

b. $2x^2 - 9x + C$

c. $4x + 9 + C$

d. $2x^2 + 11x + C$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: MCQ — Definite integrals of trigonometric and exponential functions • Cross-Lesson: 8.01 Exponentials & Logarithms

Find $\int 3\cos x\,dx$.

a. $3 - \sin x + C$

b. $4\sin x + C$

c. $3\sin x + C$

d. $3\cos x + C$

© ClipSAT — clipsat.org | All Rights Reserved

39 [Challenging] • Target: MCQ — Area under a curve • Cross-Lesson: 9.01 Statistics & Probability

Evaluate $\int_1^3 6x\,dx$.

a. 22

b. 26

c. 24

d. -24

40 [Challenging] • Target: MCQ — Signed area and regions below the axis • Cross-Lesson: 1.01 Algebra & Equations

Evaluate $\int_0^3 (9x^2 - 1)\,dx$.

a. 76

b. 78

c. 80

d. -78

41 [Challenging] • Target: MCQ — Properties of definite integrals • Cross-Lesson: 2.01 Functions & Transformations

A function satisfies $f'(x) = -2x + 5$ and $f(1) = 10$. Find $f(x)$.

a. $f(x) = x^2 + 5x + 6$

b. $f(x) = -x^2 + 5x + 9$

c. $f(x) = -x^2 + 5x - 6$

d. $f(x) = -x^2 + 5x + 6$

42 [Challenging] • Target: MCQ — Finding a function from its derivative • Cross-Lesson: 3.01 Sequences & Series

Find the area under $y = 3x^2$ from $x = 0$ to $x = 4$, using a definite integral.

a. 192

b. 66

c. 62

d. 64

43 [Challenging] • Target: MCQ — Average value of a function • Cross-Lesson: 4.01 Trigonometry

Given $\int_0^2 f(x)\,dx = 5$ and $\int_2^6 f(x)\,dx = 2$, find $\int_0^6 f(x)\,dx$.

a. 3

b. 9

c. 10

d. 7

44 [Challenging] • Target: MCQ — Word problems: integration in context • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

Given $\int_0^5 f(x)\,dx = 3$, find $\int_5^0 f(x)\,dx$.

a. 3

b. -3

c. -5

d. 5

© ClipSAT — clipsat.org | All Rights Reserved

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 6.01 Limits & Derivatives

Find the average value of $f(x) = -2x + 4$ on $[1, 4]$.

a. -1

b. -2

c. 0

d. 1

© ClipSAT — clipsat.org | All Rights Reserved